

Design Choices and Reasoning for the Knowledge Graph Extraction System

1. Overview

This project aims to develop an **end-to-end pipeline** that converts natural-language documents into a structured **Knowledge Graph (KG)**. The KG represents **entities, relations, events, and personality traits** in a machine-readable form, enabling reasoning and explainability.

Since no real annotated dataset is available, we generated **synthetic text** to evaluate and debug the pipeline. The design emphasizes *transparency, modularity, and reproducibility* rather than optimizing for raw model performance.

2. Synthetic Data Design

2.1 Why Synthetic Data

Real-world text corpora with personality annotations and ground-truth relations are scarce and costly to label.

Thus, synthetic data allows:

- **Control over narrative content** (to check correctness manually).
- **Rich personality diversity** (traits like optimism, greed, laziness).
- **Evaluability through common knowledge** (e.g., SpongeBob's characters).

2.2 What Data to Generate

We created a **1,000+-word fictional narrative** based on *SpongeBob SquarePants*.

The text includes:

- Multiple characters with known personalities.
- Interconnected social and professional relations.
- Temporal and causal events (e.g., a competition, daily routines).
- Emotional cues and dialogue reflecting personality.

2.3 Justification

The SpongeBob universe was chosen because:

- It is **universally recognizable** (enabling human domain evaluation).

- Character traits are **consistent and distinct** (good for personality modeling).
- It provides a **bounded yet diverse world** (relations like *friend_of*, *works_at*, *rival_of*, etc.).
- We can manually craft a **gold-standard JSON** for validation without subjective ambiguity.

This synthetic setup allows qualitative evaluation — comparing system outputs to human domain knowledge — without requiring labeled datasets.

3. Evaluation Approach (Conceptual)

Note: Evaluation metrics are not implemented in this iteration; only the conceptual plan is outlined.

3.1 Purpose

The evaluation aims to measure how well the extracted KG reflects the semantics of the original text.

3.2 Metrics (Conceptually)

Component	Intended Metric	Rationale
Entities	Precision / Recall / F1	Span detection correctness.
Coreference	CEAF or B ³ F1	Entity merging consistency.
Relations / Events	Triple F1	Correctness of extracted facts.
Traits	AUROC / Correlation	Trait inference alignment with expectations.
Confidence	Calibration (ECE / Brier)	Reliability of confidence values.

3.3 Justification

We rely on **domain-based correctness** — if the system identifies SpongeBob as “friendly” and “works_at Krusty Krab”, it demonstrates logical alignment.

Thus, **qualitative human evaluation** substitutes quantitative metrics for now.

This domain-knowledge-based validation is feasible because we can easily judge correctness for fictional characters.

4. Implementation Design

4.1 Multi-stage Pipeline

The pipeline is **modular and chained**, ensuring interpretability:

Stage	Description	Justification
P1 – Entities & Mentions	Identify people, organizations, locations, and artifacts.	Separates detection from reasoning; chunk-level extraction avoids token limits.
P2 – Coreference / Canonicalization	Merge duplicate mentions across chunks into canonical entities.	Maintains graph consistency; one node per real-world entity.
P3 – Relations & Events	Extract contextual relations and event structures between canonical entities.	Structured prompts with JSON enforcement ensure standardization.
P4 – Trait Evidence Extraction	Identify and quantify personality traits using contextual windows.	Provides human-like attributes with textual evidence support.
R1–R3 – Reducers / Fusion	Merge redundant facts and calibrate trait confidence.	Guarantees a clean and explainable final graph.

Each module is **independently callable**, allowing testing, caching, or replacement with improved models later.

5. Personality Representation

5.1 Chosen Representation

Each person node has associated **trait edges** of the form:

(person_id) -[HAS_TRAIT {score, confidence, evidence}]→ (trait)

Traits are expressed in **Big Five**-style or behaviorally descriptive labels (e.g., *Conscientiousness*, *Greed*, *Loyalty*, *Courage*).

5.2 Justification

- Personality is **continuous and context-dependent**, not binary.
- Representing traits as **edges** rather than node attributes allows:
 - Multiple evidence sources per trait.
 - Temporal evolution (same person can change over time).
 - Confidence aggregation from multiple documents.

5.3 Calibration

A Bayesian aggregation (Beta distribution) fuses trait evidence into a calibrated (score, confidence) pair — allowing uncertainty to be tracked.

6. LLM Workflow Piping

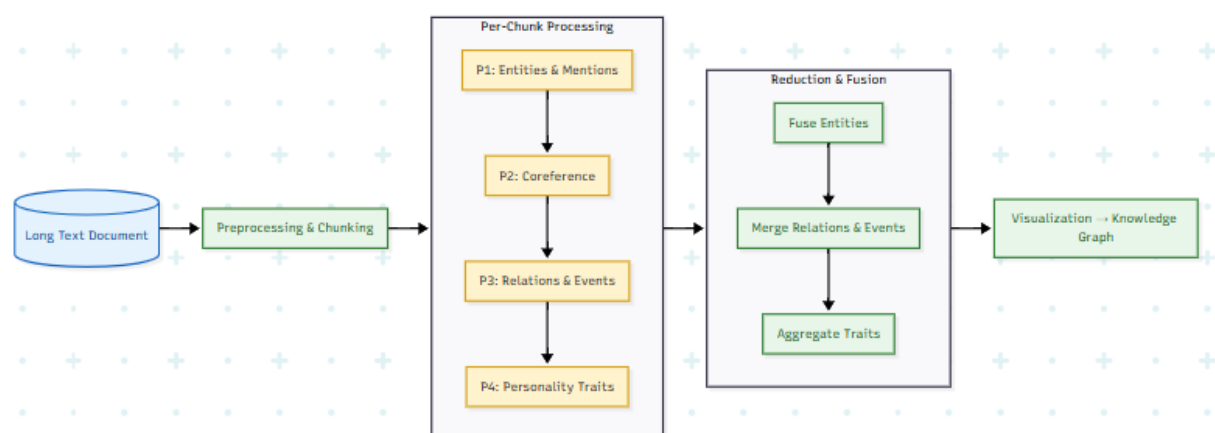
6.1 Why a Chain-of-Prompts Design

Knowledge extraction from unstructured text cannot be reliably achieved in a **single LLM prompt**, because:

- LLMs mix entity, relation, and trait reasoning.
- Prompt length and token constraints limit accuracy.
- Errors compound when everything is done in one call.

6.2 Workflow Chain

Step	Function	I/O
LLM#1	Extract entities & mentions (P1).	Input: text → Output: entity JSON.
LLM#2	Canonicalize / merge entities (P2).	Input: all entities → Output: canonical map.
LLM#3	Extract relations & events (P3).	Input: chunk text + entity map → Output: relation/event JSON.
LLM#4	Infer traits per person (P4).	Input: text windows → Output: trait probabilities & evidence.



Each stage produces structured JSON outputs, validated before feeding into the next stage. This modularity improves **fault tolerance**, **debuggability**, and **reusability** while minimizing **token cost**.

7. Data Processing and Normalization

7.1 Normalization Strategy

Aspect	Normalized	Reason
Entity Names	Lowercased + deduplicated	Ensures consistent coref matching.
Relation Types	Snake_case	Standardizes across documents (e.g., “works at” → works_at).
Trait Labels	Title Case	Easier interpretation and merging.
Spans / Indexes	Clipped to chunk boundaries	Ensures safe alignment and visualization.
Raw Text	Not lowercased globally	Keeps casing for span integrity and sentence boundaries.
Numeric Scores	Scaled [0,1]	Enables probabilistic aggregation.

7.2 Justification

- Over-normalizing text (e.g., forcing lowercase) can **break character offsets**, which are required for span alignment.
- Only semantic identifiers (like relation types) need normalization for consistency across LLM outputs.
- Maintaining case and punctuation ensures that the **LLM retains natural cues** for emotion and tone — crucial for personality modeling.

8. Insights & Limitations

8.1 Insights

- **Structured prompting** significantly improves reliability over free-form extraction.
- **SpongeBob-based synthetic data** allows immediate qualitative evaluation using human intuition.
- Modeling **personality as evidence-backed edges** provides interpretability and psychological realism.
- The **multi-prompt LLM chain** balances precision and cost by focusing each prompt on one reasoning task.

8.2 Limitations

- **LLM hallucination risk:** Even structured prompts may yield non-existent relations.
- **Subjectivity in traits:** Personality inference depends on textual cues; no ground-truth ratings exist.
- **Scalability:** Each pipeline stage invokes the LLM multiple times, leading to latency and cost.
- **No quantitative evaluation yet:** Metrics (F1, AUROC) planned for future implementation.
- **Synthetic bias:** Fictional data may not generalize to real-world domains.

9. Conclusion

The project demonstrates a **modular, interpretable approach** to Knowledge Graph construction from text using chained LLM workflows.

By using **synthetic narrative data** with recognizable characters, we can intuitively validate correctness and behavior without large-scale annotation.

This design prioritizes *clarity, structure, and explainability* — making it a strong foundation for future work in automated knowledge extraction and personality-aware AI systems.

10. Generative AI

The report and accompanying code were developed with significant assistance from Generative AI, specifically ChatGPT by OpenAI. The AI was used to generate, refine, and debug code, as well as to support the writing and structuring of this report.

A record of the session can be accessed here: <https://chatgpt.com/share/68f376ec-68d8-800f-9c5d-3b7c39d5ae19>